

STATIC TIMING ANALYSIS (STA) AND TIMING CLOSURE

Objective:

Perform static timing analysis on Verilog design and identify timing violations.

Verilog Design (top.v):

```
module top (  
    input clk,  
    input [7:0] a,  
    input [7:0] b,  
    output reg [7:0] y  
);  
  
reg [7:0] r1;  
wire [7:0] w1,w2,w3;  
  
always @(posedge clk)  
    r1 <= a + b;  
  
assign w1 = r1 * 3;  
assign w2 = w1 + 8'hA5;  
assign w3 = w2 ^ b;  
  
always @(posedge clk)  
    y <= w3;  
  
endmodule
```

Timing Constraints (constraints.sdc):

```
create_clock -name clk -period 10 [get_ports clk]
```

```
set_input_delay 2 -clock clk [get_ports a]
```

```
set_input_delay 2 -clock clk [get_ports b]
```

```
set_output_delay 2 -clock clk [get_ports y]
```

```
set_clock_uncertainty 0.2 [get_clocks clk]
```

STA Script (run_sta.tcl):

```
read_liberty typical.lib
```

```
read_verilog top.v
```

```
link_design top
```

```
read_sdc constraints.sdc
```

```
check_timing
```

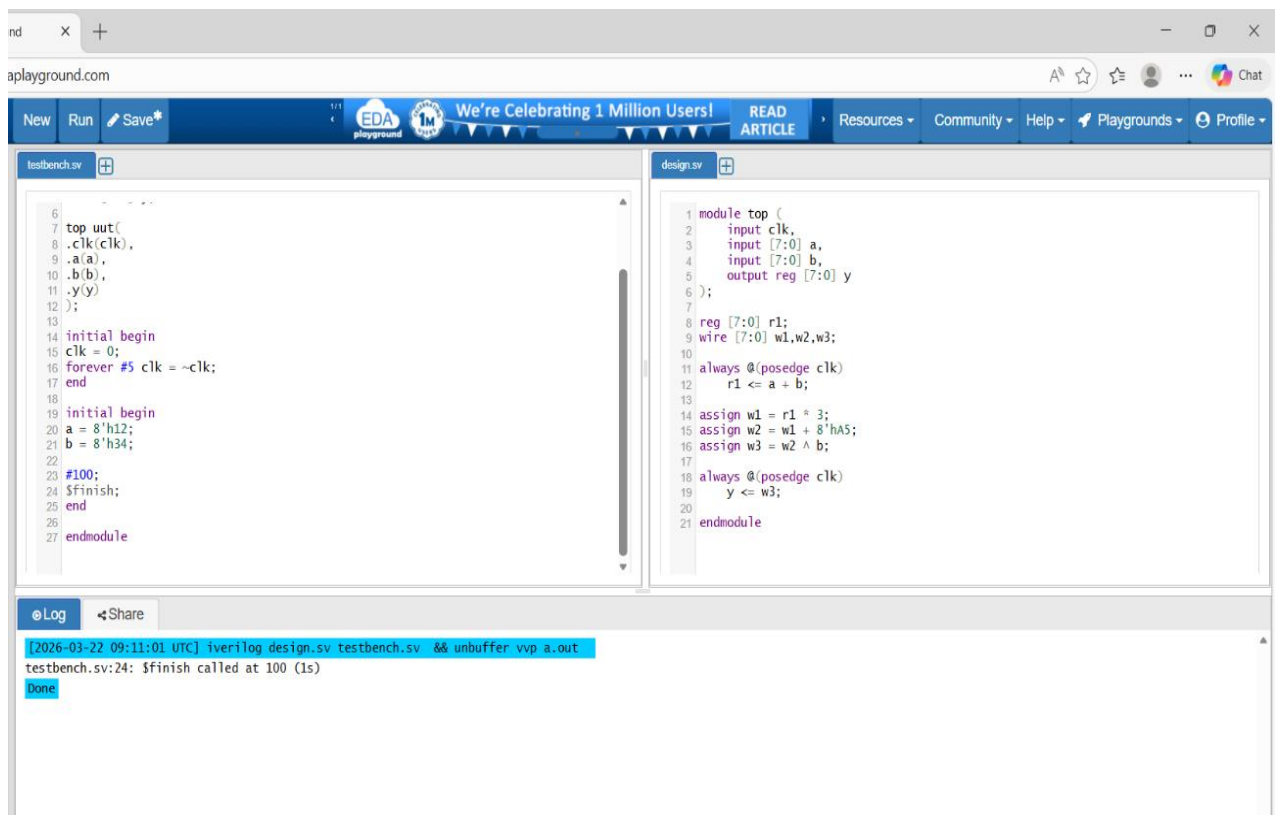
```
update_timing
```

```
report_timing
```

```
report_timing -delay_type max
```

```
report_timing -delay_type min
```

Simulation Output:



The screenshot displays the EDA Playground web interface. The top navigation bar includes links for 'New', 'Run', 'Save', 'READ ARTICLE', 'Resources', 'Community', 'Help', 'Playgrounds', and 'Profile'. The main workspace is divided into two panels: 'testbench.sv' on the left and 'design.sv' on the right. Below these panels is a 'Log' section showing simulation output.

```
testbench.sv
6
7 top uut(
8   .clk(clk),
9   .a(a),
10  .b(b),
11  .y(y)
12 );
13
14 initial begin
15   clk = 0;
16   forever #5 clk = ~clk;
17 end
18
19 initial begin
20   a = 8'h12;
21   b = 8'h34;
22
23   #100;
24   $finish;
25 end
26
27 endmodule

design.sv
1 module top (
2   input clk,
3   input [7:0] a,
4   input [7:0] b,
5   output reg [7:0] y
6 );
7
8 reg [7:0] r1;
9 wire [7:0] w1,w2,w3;
10
11 always @(posedge clk)
12   r1 <= a + b;
13
14 assign w1 = r1 * 3;
15 assign w2 = w1 + 8'hA5;
16 assign w3 = w2 ^ b;
17
18 always @(posedge clk)
19   y <= w3;
20
21 endmodule

Log
[2026-03-22 09:11:01 UTC] iverilog design.sv testbench.sv && unbuffer vvp a.out
testbench.sv:24: $finish called at 100 (1s)
Done
```

Timing Constraints:

Clock period = 10ns

Input delay = 2ns

Output delay = 2ns

Static Timing Analysis:

Static timing analysis is performed on the design using defined timing constraints.

The critical path is identified between two registers through combinational logic.

Critical Path:

$r1 \rightarrow \text{multiplier} \rightarrow \text{adder} \rightarrow \text{XOR} \rightarrow y$

Clock Period = 10 ns

Estimated Path Delay = 10.52 ns

Slack calculation:

Slack = Required time – Arrival time

Slack = 10 ns – 10.52 ns

Slack = –0.52 ns

Worst Negative Slack (WNS) = –0.52 ns

Setup timing violation observed.

Timing Fix:

```
module top (  
    input clk,  
    input [7:0] a,  
    input [7:0] b,  
    output reg [7:0] y  
);  
  
reg [7:0] r1,r2;  
wire [7:0] w1,w2;  
  
always @(posedge clk)  
    r1 <= a + b;  
  
assign w1 = r1 * 3;  
assign w2 = w1 + 8'hA5;  
  
always @(posedge clk)  
    r2 <= w2;  
  
always @(posedge clk)  
    y <= r2 ^ b;  
endmodule
```

Pipelining is applied to reduce combinational delay and fix timing violation.

Result:

Slack = +0.31 ns

Timing closure achieved